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Nitrogen-fixing bacteria boost floral attractiveness in a tropical legume species during nutrient limitation

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Abstract

Premise: Legumes establish mutualistic interactions with pollinators and nitrogen (N)-fixing bacteria that are critical for plant reproduction and ecosystem functioning. However, we know little about how N-fixing bacteria and soil nutrient availability affect plant attractiveness to pollinators.

Methods: In a two-factorial greenhouse experiment to assess the impact of N-fixing bacteria and soil types on floral traits and attractiveness to pollinators in *Chamaecrista latistipula* (Fabaceae), plants were inoculated with N-fixing bacteria (NF+) or not (NF-) and grown in N-rich organic soil (+N organic soil) or N-poor sand soil (-N sand soil). We counted buds and flowers and measured plant size during the experiment. We also measured leaf, petal, and anther reflectance with a spectrophotometer and analyzed reflectance curves. Using the bee hexagon model, we estimated chromatic contrasts, a crucial visual cues for attracting bees that are nearby and more distant.

Results: NF+ plants in -N sand soil had a high floral display and color contrasts. On the other hand, NF- plants and/or plants in +N organic soil had severely reduced floral display and color contrasts, decreasing floral attractiveness to bee pollinators.

Conclusions: Our findings indicate that the N-fixing bacteria positively impact pollination, particularly when nutrients are limited. This study provides insights into the dynamics of plant–pollinator interactions and underscores the significant influence of root symbionts on key floral traits within tropical ecosystems. These results contribute to understanding the mechanisms governing mutualisms and their consequences for plant fitness and ecological dynamics.

KEYWORDS

belowground interactions, *Chamaecrista*, mutualistic interactions, nitrogen fixation, pollinator attraction, tri-trophic interaction

Plants often engage in simultaneous mutualistic relationships with multiple partners (Bronstein, 1994; Mack and Rudgers, 2008; Afkhami et al., 2014), both above- and belowground, that impart distinct benefits to plants (Hoeksma and Bruna, 2015). While many studies have traditionally focused on the interference of aboveground mutualisms on each other (e.g., Godschalx et al., 2015; Melati and Leal, 2018), the presence of belowground mutualists can influence the interactions of plants with aboveground mutualistic partners (Bardgett and Wardle, 2003; Afkhami et al., 2020; Keeler et al., 2021). For instance, the mutualistic symbiosis between plants and mycorrhizal fungi commonly positively impacts floral production and size (Gange and Smith, 2005; Perner et al., 2007; Varga and Kytöviita, 2010), with positive consequences for pollinator visits to flowers (Gange and Smith, 2005; Appendix S1). Conversely, neutral and negative impacts of belowground mutualisms on aboveground mutualisms have been reported (Table 1). For example, N-fixing bacteria can reduce extrafloral nectar production, decreasing ant attraction and potential herbivore protection (e.g., Godschalx et al., 2015). This interplay between below- and aboveground mutualisms highlights the necessity for an integrated understanding of these relationships because they collectively shape the functioning and dynamics of terrestrial ecosystems (Barber and Soper Gorden, 2015; Magnoli et al., 2023).

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TABLE 1 Summary of studies on the effect of belowground mutualists (nitrogen-fixing bacteria and mycorrhiza) on floral traits and/or pollination. For the complete list of reviewed studies, see Appendix S1.

		Number of studies (%)			Number of plant species (%)		
Belowground mutualist	**Plant life form**	**Crop/ornamental plants**	**Noncrop plants**	**Floral traits measured across studies[a]**	**Only one floral trait measured (%)[b]**	**Effect of belowground mutualists on floral traits (% of positive effect detected)[c]**	**Effect of belowground mutualists on pollination (% of positive effect detected)[c]**
Mycorrhiza (31 studies)	Annual	7 (16)	5 (12)	Inflorescence number, inflorescence size, flower number, flower diameter, petal size, nectar amount and concentration, pollen production per flower, ovary mass.	8 (67)	12 (58)	8 (75)
	Perennial	15 (35)	12 (28)	Number of inflorescences and inflorescence buds, inflorescence mass, number of flowers and flower buds, flower diameter, number of functional stamens, number of pollen grains, sugar mass in floral nectar, number of flowering plants, time to floral anthesis.	19 (53)	36 (61)	12 (25)
N-fixing bacteria (4 studies)	Annual	2 (5)	1 (2)	Flower number per inflorescence, flower mass, pollen count per flower	2 (100)	2 (0)	2 (100)
	Perennial	1 (2)	0 (0)[d]	Flower number per plant	2 (100)	2 (100)	2 (100)

Notes: We searched for papers in the *Web of Science* database (http://www.webofknowledge.com/) for all available years up to December 2023. Key search terms: "nitrogen fixing bacteria" OR "nitrogen-fixing bacteria" OR rhizobi* OR nodul* OR mycorrhiz*) AND (flower* OR "flower trait*" OR "flower attractiveness" OR "floral trait*" OR "floral" OR inflorescenc* OR "flower phenolog*" OR "floral phenolog*" OR "flowering phenolog*" OR pollination OR "pollen transport") AND (pollinato* OR mutualism* OR mutualist* OR "tritrophic interaction*" OR "higher-order interaction*" OR tripartite). The search resulted in 209 papers. We read titles and abstracts of all papers and included here 35 studies in which the effect of belowground plant mutualists was evaluated on floral descriptors and/or pollination.

[a]Studies did not use the same floral traits, making comparisons across studies difficult, with almost all of them not including descriptors of flower phenology and color.

[b]Most studies used only one floral trait, which, in general, is the number of inflorescences or flowers.

[c]The effect of mycorrhiza and rhizobium on floral traits and pollination was predominantly positive, although mycorrhizas caused substantial neutral or negative effects.

[d]Only our study, including multiple floral traits measured, is in this category.

Nitrogen-fixing bacteria are crucial as belowground mutualists, especially with legume species (Sprent, 2001). This mutualistic partnership unfolds within specialized root structures called nodules, where bacteria fix N and supply it to plants and obtain plant-derived carbon compounds (Phillips, 1980; Tate, 1995; Dilworth et al., 2008). Legumes have various strategies to interact with N-fixing bacteria, depending on the environmental context (Menge et al., 2009, 2015). For instance, plants in tropical environments tend to have a facultative strategy with N-fixing bacteria. This strategy is suggested to arise in tropical forest soils because they are typically N-rich (Hedin et al., 2009), favoring plants that regulate their N-fixation rate also using the available soil N (Menge et al., 2015). In contrast, plants with an obligate interaction strategy with N-fixing bacteria maintain a consistent N-fixation rate, regardless of soil N content (Menge et al., 2015). Considering the frequent N constraints in soil (Ågren et al., 2012), this N-fixing ability emerges as a valuable N source, promoting plant growth, development, and overall fitness (Walker et al., 2012), which may consequently impact aboveground mutualisms.

Many legume species depend on N-fixing bacteria to obtain N resources for their growth and reproduction (Mathesius, 2022) while relying on pollinators for sexual reproduction (Suso et al., 2016). While these interactions are physically separate, they are interconnected through plant resource allocation, a process mainly influenced by the availability of N fixed by bacteria (Godschalx et al., 2015). The N-fixing capacity of the bacteria in the root nodules directly influences the plant's N content (Lindström and Mousavi, 2020). This influence, in turn, establishes a positive feedback loop, enhancing photosynthesis, accelerating vegetative growth rates, and bolstering energy reserves (Dilworth et al., 2008). Consequently, plants interacting with N-fixing bacteria increase floral production, becoming more attractive to pollinators (e.g., Pereira et al., 2019). However, only four studies have directly investigated the impact of N-fixing bacteria on floral attractiveness to pollinators, in contrast to 31 studies on the impact of mycorrhizae (see Table 1). Our understanding of the positive impact of soil N on floral features is more robust. Plant uptake of N from the soil leads to significant changes in aboveground plant traits, resulting in increased floral production, extended floral phenology, and enhanced floral color attractiveness (Lau and Stephenson, 1993; Wagner, 2012; Ruamrungsri et al., 2021). In contrast, the effects of N-fixing bacteria can be considerably complex, with multiple positive effects beyond uptake of the fixed N (e.g., García-Fraile et al., 2012; Allito et al., 2020). The presence or absence of N-fixing bacteria can influence plant–pollinator interactions by modifying various floral attributes and phenology (Keeler et al., 2021), especially in plants that rely heavily on bacteria for N and form root nodules.

Mutualistic interactions are inherently shaped by the ecological conditions (Chamberlain et al., 2014). The conditional nature of mutualisms implies that different ecological factors, including nutrient availability, can significantly mold the dynamics of plant mutualisms (Bronstein, 2015), including the establishment and maintenance of both nutritional and pollination mutualisms. For example, the effect of soil nutrient availability on floral attractiveness and pollination can vary widely from positive (Muñoz et al., 2005), neutral (Gijbels et al., 2015) to even negative on floral production (Vaudo et al., 2022). However, changes in floral attributes are contingent on soil nutrient levels and species-specific traits (David et al., 2019). For some plants, changes in soil features can directly impact the floral traits (e.g., Klein et al., 2015), but for legumes, these changes can also modify the intensity of the plant's interactions with N-fixing bacteria (see Sepp et al., 2023 and references therein), which, in turn, could also modify floral traits (Pereira et al., 2019). Therefore, direct and indirect effects on plant nutrition can change the timing and efficiency of pollination, thus influencing plant reproductive success (Vaudo et al., 2022). Consequently, understanding how specific soil types and N-fixing bacteria influence floral attractiveness to pollinators is fundamental for a comprehensive understanding of ecosystem dynamics (David et al., 2019).

Chamaecrista species (Fabaceae) establish mutualistic interactions with N-fixing *Bradyrhizobium* bacteria (Sprent et al., 2017; Casaes et al., 2024) and are commonly distributed in sand soils (e.g., Fidalgo et al., 2018). Some *Chamaecrista* species produce nodules with symbiosomes (de Faria et al., 2022; Casaes et al., 2024), organelles occupied by bacteroids with high compartmentalization and integration between bacteria and plant cell ultrastructure, which are supposed to be more efficient and stable (Mohd-Radzman and Drapek, 2023). Using this system, we investigated the knowledge gap regarding how these bacteria and soil nutrient levels impact floral attributes related to pollination. We conducted a multifactorial controlled greenhouse experiment with *Chamaecrista latistipula* (Benth.) Afr. Fern. & E. Nunes (Fabaceae), in which the presence of these N-fixing bacteria and soil features were manipulated. During the 16-month experiment, we measured the constancy and intensity of bud and floral production during plant development. Additionally, we measured the reflectance of leaves, petals, and anthers using a spectrophotometer under varying treatment conditions and analyzed the reflectance curves. We postulated that the N-fixing bacteria increase floral production, constancy, and the visual attractiveness of plants to bee pollinators. In contrast, the absence of N-fixing bacteria may adversely impact plant attractiveness unless mitigated by N-rich soils. In this complementary scenario, some soil types might provide an alternative N source for plants, offsetting the absence of N-fixing bacteria.

MATERIALS AND METHODS

Study system

Our focal plant species, *Chamaecrista latistipula* (Benth.) Afr.Fern. & E.P. Nunes, is widely distributed across Brazil and is pollinated by vibrating bee species (see Nogueira

et al., 2018—in which this species is referred to as *Chamaecrista desvauxii* var. *latistipula*). This species is a shrub with a root system with a high number of nodules (Figure 1) containing N-fixing bacteria (see Naisbitt et al., 1992 for the anatomical characterization of root nodules of *Chamaecrista desvauxii* var. *latistipula*), previously characterized as *Bradyrhizobium* species (Casaes et al., 2024). The compound leaves of *C. latistipula* have four distinct leaflets, and the petiole bears an extrafloral nectary.

This species produces cymose inflorescences containing one to three asymmetric pentamerous yellow flowers. The flower comprises five distinct whorls: a solitary whorl of sepals and petals, two whorls of fertile stamens with poricidal anthers, and a single-carpel gynoecium that encloses ovules (Nogueira et al., 2018; Figure 1B). The flowers offer only pollen as a resource to pollinators (Faegri and van der Pijl, 1979; Barbosa et al., 2023). Anthers dehisce through apical pores, releasing pollen grains when bees

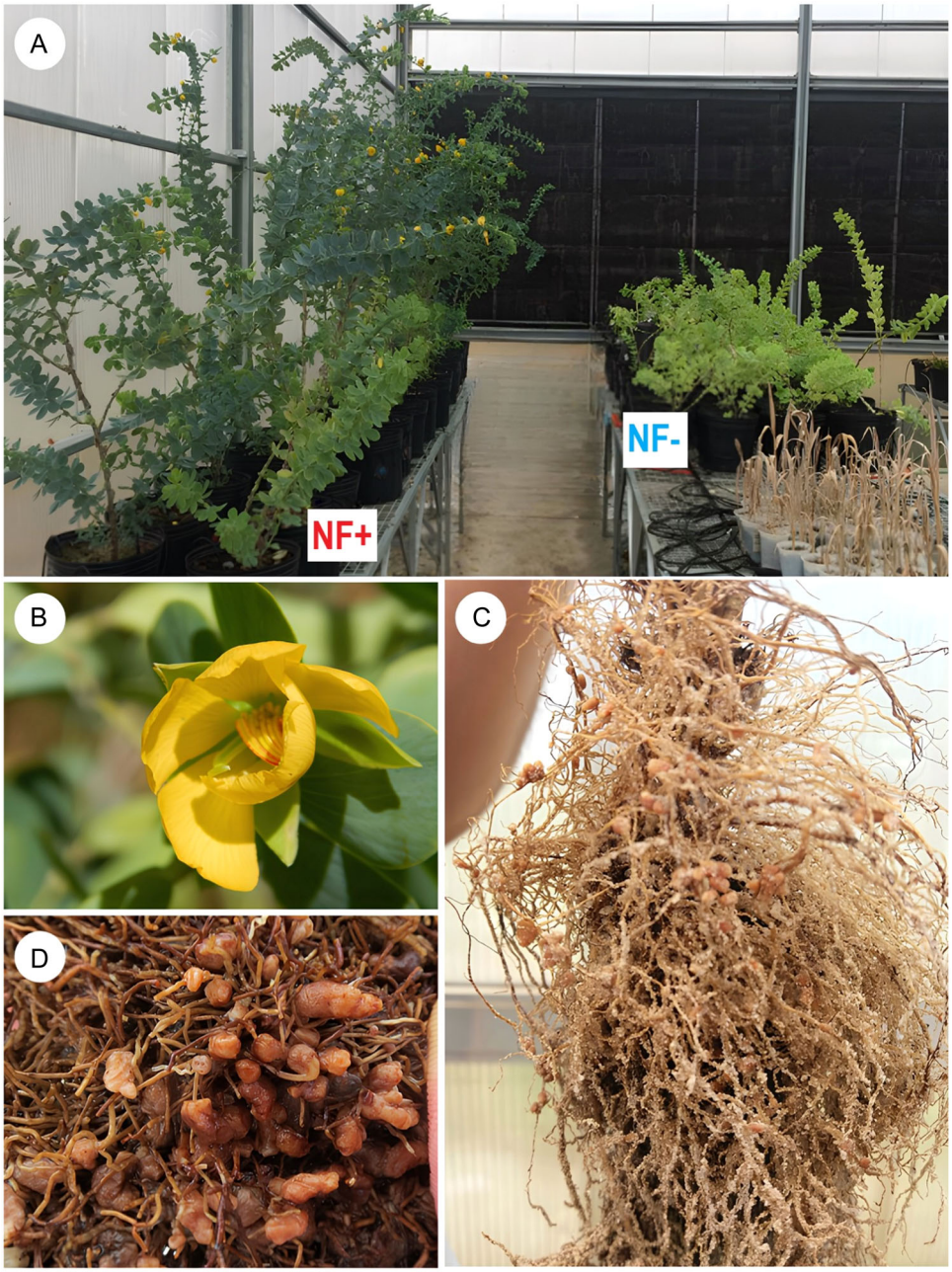


FIGURE 1 Plants, flower, and roots of *Chamaecrista latistipula* at the end of the greenhouse experiment. (A) Plants in N-poor sand soil or N-rich organic soil with N-fixing bacteria (NF+ in red, left) or without (NF- in blue, right). Among the NF+ plants, the tall plants with multiple flowers were grown in N-poor sand soil; the shorter plants with yellowish green leaves were grown in N-rich organic soil. All NF- plants, regardless of soil type, were shorter with chlorotic leaves. (B) Open flower on NF+ plant in the N-poor sand soil (control). (C) Root system of a control plant with some nodules and sand particles adhering to larger and thinner roots. (D) Washed roots and nodules from a control plant. All plants were 1-year-old when planted in the soils.

vibrate flowers during buzz pollination (Buchmann, 1983; Gottsberger and Silberbauer-Gottsberger, 1988). Anthesis is limited to the early hours of the day, typically from 06:30 to 14:00 hours, with variations observed among different populations (Nogueira et al., 2018). The fruits are linear, flat legumes with two valves that undergo explosive dehiscence (Irwin and Barneby, 1982).

Seed planting and experimental conditions

Considering that the mutualistic root–symbiont interaction varies depending on soil conditions and the availability of inorganic soil N (Weese et al., 2015), we designed a 2×2 factorial experiment in which we manipulated the N-fixing bacteria and soil characteristics. Initially, we harvested fruits from 10 marked mother plants in a natural area in Botucatu, São Paulo, Brazil (22°53′S 48°29′W). In September 2017, from each mother plant, we harvested two to four mature fruits that were close to dehiscing. Fruits were chosen based on being fully developed, dry, with a dark color, undamaged, and containing a substantial quantity of undamaged healthy seeds. For seed planting, we sterilized seeds and 60 clean pots (30 L) sequentially in 70% v/v ethanol, 5% v/v sodium hypochlorite, and 3% v/v hydrogen peroxide (Somasegaran and Hoben, 1994). Seeds were then rinsed with distilled water before mechanically scarifying them to remove primary dormancy.

The experiment included two types of soils: (1) A N-poor soil (0.5 g of N per 1000 g of soil) consisted primarily of sand (90% sand mixed with 10% topsoil) with low nutrient concentration, particularly N (0.5 g of N per 1000 g of soil) (hereafter, -N sand soil); and (2) a N-rich organic soil (5.05 g of N per 1000 g of soil) composed of 100% topsoil, to which 100 mL of 2 M KNO_3 buffer solution was added per pot after 45 days of seed planting (hereafter, +N organic soil).

Before beginning the experiment, we autoclaved the substrate to sterilize it. Finally, we planted four seeds in each pot. For the bacterial inoculation, fresh nodules were collected from the root system of three *C. latistipula* plants in the field, then freshly crushed root nodules were immersed in 1 L of distilled water, and poured on the soil at 1 and 2 weeks after seed planting. The pH of both substrates was close to neutral at the beginning of the experiment, and we monitored the pH during the first 6 months of the experiment because of its influence on root–bacteria associations (Thomas et al., 1997). The pH did not change greatly over time.

The experiment involved four inoculum–soil treatments with N-fixing bacteria (NF+) or without (NF-) the bacteria in N-poor or N-rich soil: (1) NF+ and -N sand soil (control); (2) NF- and -N sand soil; (3) NF+ and +N organic soil; and (4) NF- and +N organic soil. The biotic and abiotic factors within the Van der Hoeven (Den Hoorn, The Netherlands) greenhouse at the Federal University of ABC were rigorously controlled. A climate control system maintained mild relative humidity and temperature, and an automated drip irrigation system supplied 150 mL of water at 06:00 and 18:00 hours daily. Cross-contamination between experimental groups was prevented by regularly sanitizing greenhouse benches with 70% v/v ethanol and 3% v/v hydrogen peroxide.

Assessment of plant attractiveness traits to pollinators

Every month, from September 2017 to January 2019, we systematically counted the leaves and leaf scars and measured the height for all plants. Starting in October when plants began producing the initial flower buds, every morning we counted all floral buds and the open flowers (daily floral display) and calculated the cumulative floral display (total flowers per plant over 3 days).

Since plant size can impact floral attractiveness (e.g., Lortie and Aarssen, 1999), we chose two different descriptors of plant size that could indirectly modify pollinator attraction: plant height (cm) and the number of leaf scars and leaves on the branches. The leaf scars of this species persist along the entire stem, including the basal and older portions of plants. The sum of the leaf scars and leaves per plant (hereafter, plant size) represents the aboveground plant body, even when the plant has multiple stems and branches (see similar evaluation in Baker-Méio and Marquis, 2012). In our case, when the plant size or height is greater, we expect the flowers to be more visually exposed to floral visitors, especially bees, making these plants more attractive to pollinators.

Similarly, we used the number of flower buds and the daily and cumulative floral display as descriptors of plant attractiveness to pollinators. In this case, the greater the availability of flowers and future production of flowers (represented by flower buds), the higher the plant's attractiveness to pollinators (Klinkhamer and de Jong, 1993). We also calculated floral density per plant on each day (total number of flowers divided by the size of the plant on that day) to control for plant size when assessing floral production.

We also investigated the effect of N-fixing bacteria and soil type on flowering phenology, given that the predictability and synchronicity of floral production within and among plants can enhance or diminish the attractiveness of individual plants or the plant population to pollinators (Lázaro et al., 2020). To assess flowering phenology, we recorded the number of days from seed germination until the first flowers opened and the maximum number of flowers produced per plant over time in each experimental plant group.

Finally, we were also interested in visual floral signals that could modify floral attractiveness. Given that flowers of *Chamaecrista* species have UV-reflective petals and UV-absorbing anthers (Tunes et al., 2021), we measured reflectance from the central region of the larger petal (cucculus), the short anthers, and the long anthers of each flower after the

plants had grown for 1 year. Stamens in these flowers have significantly reduced filaments and consist almost entirely of anthers (Figure 1B) that have an apical pore (Nogueira et al., 2018). For each flower sampled, we combined three to five anthers of each size for reflectance measurements due to the narrow width of isolated anthers. We measured 21 freshly opened flowers (~96% from different individuals) using a properly calibrated USB4000 reflectance spectrophotometer (OceanOptics, Dunedin, FL, USA). Of these 21 flowers, 12 were from NF+ plants in -N sand soil (controls, field condition). Since the other treatments produced far fewer flowers, especially toward the end of the experiment, we sampled four flowers (total from three plants) from NF+ plants in +N organic soil and five (total from five plants) in NF- plants in +N organic soil. In the last treatment, NF- plants in -N sand soil rarely produced flowers (only one during our sampling), and we did not obtain flowers for reflectance measurements. We used barium sulfate and a black chamber as white and black standards, respectively (Lunau et al., 2011). We took all reflectance measurements at a 45° angle using a UV/SR-VIS reflectance probe (Stellarnet, Tampa, FL, USA) from 200 to 1100 nm; however, for the analyses, we only used measurements from 300 to 700 nm, the spectrum visible to pollinators.

Using the hexagon color vision model, we obtained color variables in the subjective bee vision (Chittka, 1992). We used the spectral sensitivities of the bumblebee *Bombus terrestris* as the bee model (Skorupski et al., 2007). This choice is based on the effectiveness of large bees as primary pollinators of plant species within the Cassinae tribe (e.g., Saab et al., 2021), including our focal species (Barbosa et al., 2023). We used the standard solar light (D65) available in the R package pavo (Maia et al., 2013) and reflectance measurements of the plant's green leaves as a background in our model calculations. We used green leaves as a background for petals, while we considered the reflectance of the central region of the petals as the background for anthers (following Saab et al., 2021). We first calculated hyperbolically transformed quantum catches with von Kries correction using the vismodel function in pavo (Maia et al., 2013). Then, the color loci were plotted on the hexagon using the colspace function (Maia et al., 2013). We calculated green contrasts as the specific contrast produced in the green photoreceptor minus a constant of 0.5 (Spaethe et al., 2001). If the obtained contrast value was negative, we converted the value to zero. Green contrast is an important descriptor for long-distance bee attraction (Giurfa et al., 1996). Chromatic contrast with the relevant background (i.e., green leaves as the background for the petals and petal color for the stamens) is related to short-distance bee attraction (Chittka, 1992; Renoult et al., 2017; van der Kooi et al., 2018). Chromatic contrast was calculated as the distance between each color location and the achromatic center, representing the background locus (Chittka, 1992).

At 16 months after seed germination, we destructively sampled roots to check for root nodules and validate the experimental manipulation of N-fixing bacteria. We found nodules only on NF+ plants, indicating that our inoculum treatment and the exclusion treatment were successful. At the end of the experiment, we removed plants from the soil, washed the roots, and successively passed the sand through sieves of different sizes, collecting nodules with each pass. Subsequently, we tallied the number of nodules produced by each NF+ plant in -N sand soil and in +N organic soil. Most of these nodules were healthy-looking, completely turgid, with a distinctive pink color that indicated N fixation (Baird and Webster, 1982).

Data analyses

To evaluate the impact of the bacteria and soil type on plant height (cm) and size, we employed two linear mixed models (LMMs). In both instances, we incorporated the interaction term to assess whether the influence of N-fixing bacterial (NF+ vs. NF-), might be contingent on soil type (-N sand soil vs. +N organic soil). The models also included maternal plants and plant identity as random variables. We used 15 maternal plants, each providing seeds for all four treatments in the experiment ($N = 60$ plants). Plant identity was included in the models as a random variable due to our experimental design, which involved repeated measurements of the same plants throughout the experiment.

To examine the interactive effect of bacteria and soil type on the number of flower buds, the number of flowers opened in a single day (daily floral display), and the total number of flowers over 3 days (cumulative floral display), we used three generalized linear mixed models (GLMMs) with a negative binomial distribution for the three count variables. In these models, we included the interaction term between the bacteria and soil type to test whether the bacterial effect depended on soil type. Additionally, we included maternal plants and plant identity as random variables. We also constructed an additional model to analyze the effect of bacteria and soil type on floral density (continuous variable). In this case, we used an LMM with a Gaussian distribution and applied a log + 1 transformation to the floral density before analysis. We only included one measurement per plant (total floral density) without repeated measures over time, so we included only the maternal plants as a random variable and did not incorporate plant identity.

To evaluate the influence of bacteria and soil type on flower phenology (i.e., flower onset and flower peak), we employed two GLMMs with a negative binomial distribution. Flower onset was defined as the number of days from seed germination to the emergence of the first flower in each plant (count variable). The flower peak was described as the number of days from seed germination to the day with the most flowers (count variable). The first model used the flower onset (in days) as the response variable, while the second used the flowering peak (in days). In these two models, we included the interaction term to evaluate the interactive effect of N-fixing bacteria and soil type on flowering phenology. We also included maternal plants as a random term in both models.

Finally, we constructed four additional general linear models (LMs) to analyze the effect of N-fixing bacteria and

soil type on floral color attractiveness. We used green contrast (petal), green contrast (anther), chromatic contrast (petal), and chromatic contrast (anthers) as response variables. In those cases, we did not have flowers to estimate the contrasts of petals and anthers in the NF- plants in the sand soil, so applying a model with two fixed factors and their interaction, as previously used for other flowering descriptors, was impossible. Therefore, in these particular cases, we applied an LM with a single categorical factor to test the direct difference between the three available conditions (NF+ plants in -N sand soil, NF+ plants in +N organic soil, NF- plants in +N organic soil) on each response variable (green and chromatic contrasts for each flower part), using the plants with N-fixing bacteria in sand soil as the control (natural field condition).

In the last stage of our experiment, we tested the effect different soil types on the production of root nodules in NF+ plants. For that, we applied a GLMM with a negative binomial distribution, in which the total number of nodules was used as the response variable and the maternal plants as a random variable. Since an interaction between the addition of the bacteria and soil type was not detected in nodule formation, we ultimately employed an additive model to explore the effect of inoculation and soil type.

In all analyses, we examined residual dispersion and, if required, evaluated overdispersion. All analyses were done using R version 4.2.3 (R Core Team, 2023) with R packages glmmTMB (Brooks et al., 2017), lme4 (Bates et al., 2015), car (Fox and Weisberg, 2019), or lmerTest (Kuznetsova et al., 2017). *P*-values were obtained by applying the summary() function to the models generated by the glmmTMB and lme4 packages.

RESULTS

Only *C. latistipula* plants inoculated with N-fixing bacteria (NF+) produced root nodules (Appendix S2), confirming the effectiveness of our manipulative experiment. Among the NF+ plants, those in -N sand soil produced an average of 2776 ± 974.6 (mean ± SD) nodules per plant, 5500% more nodules than NF+ plants in +N organic soil, which produced an average of 49.1 ± 55.7 nodules per plant. This pattern indicates that root nodulation depended on the soil type (Figure 1C, D; df = 18, $Z = 24.4$, $P < 0.001$, Appendix S1). Of the 60 plants used in our experiment, four NF- plants in +N organic soil died before the end of the experiment.

Influence of N-fixing bacteria and soil type on pollinator attractiveness: plant size and floral production

We detected an interaction between the bacteria and soil type (Figure 2); the presence of the bacteria increased plant height and size, but effects were dependent upon the substrate (plant height: df = 51, $F = 65.1$, $P < 0.001$; Figure 2A; plant size: df = 38.9, $F = 191.7$, $P < 0.001$; Figure 2B). In -N sand soil, NF+ plants were approximately 1.8 times taller and 2.9 times larger than NF+ plants in +N organic soil. Conversely, NF- plants in both soil types were relatively shorter and smaller than NF+ plants.

Similarly, for almost all descriptors of floral production, we detected an interaction between the presence of the bacteria and soil type (Figure 3). The presence of the bacteria increased the production of flower buds, but this effect depended on the soil type (df = 387, $Z = 7.9$, $P < 0.001$; Figure 3A). In -N sand soil, NF+ plants produced, on average, 38 times more flower buds than NF+ plants in +N organic soil. On the other hand, NF- plants in -N sand soil produced significantly fewer flower buds than NF- plants in +N organic soil. However, we did not observe an interaction between the presence of N-fixing bacteria and the soil type for daily floral display (df = 163, $Z = 0.001$, $P = 0.99$; Figure 3B). Consequently, the impact of the bacteria and the substrate on the number of open flowers was additive

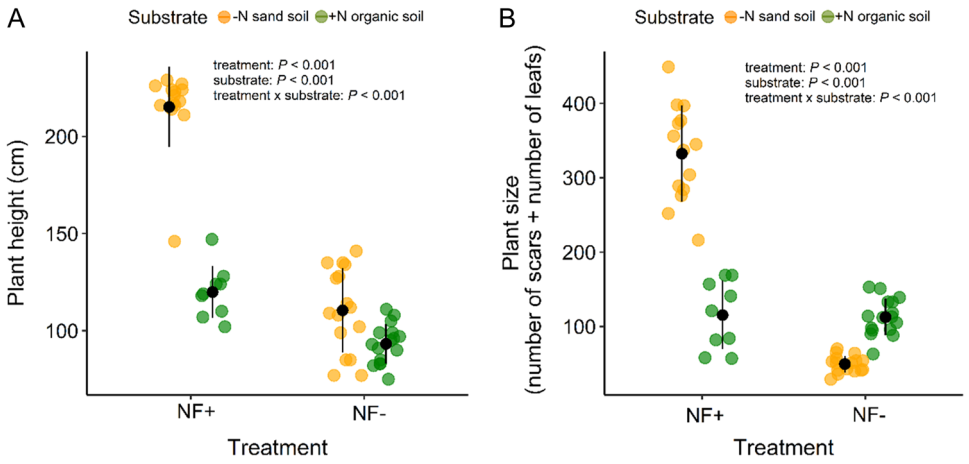


FIGURE 2 Plant responses in N-poor (-N) sand soil or N-rich (+N) organic soil with (NF+) and without (NF-) N-fixing bacteria. (A) Mean plant height and (B) plant size (combined total of leaves and scars) for each treatment combination. *P*-values were derived from general linear mixed models, reflecting the statistical significance of bacterial treatments, soil types, and their interaction on plant growth metrics.

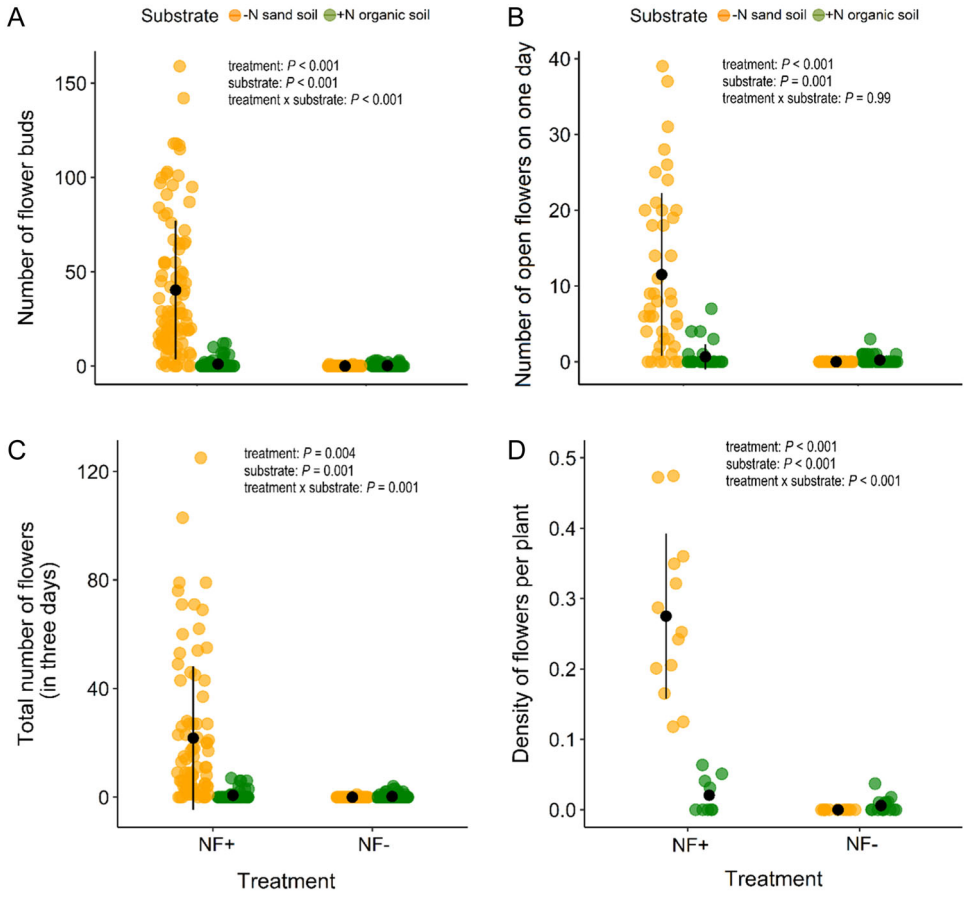


FIGURE 3 Floral traits on *Chamaecrista latistipula* plants with (NF+) and without (NF-) nitrogen-fixing bacteria in N-poor (-N) sand soil or N-rich (+N) organic soil. (A) Number of flower buds generated. (B) Daily count of open flowers. (C) Total number of flowers produced over 3 days. (D) Floral density per plant (number of flowers relative to plant size). *P*-values were derived from generalized linear mixed models for A–C and from linear mixed models for D, reflecting the statistical significance of bacterial treatments, soil types, and their interaction on floral trait metrics.

(bacteria: df = 163, $Z = 7.218$, $P < 0.001$; soil type: df = 163, $Z = 3.115$, $P = 0.001$). Specifically, on average, NF+ plants in -N sand soil produced 17.4 times more flowers per day than NF+ plants in +N organic soil. In contrast, within the NF- group, those in -N sand soil did not have any open flowers, while NF- plants in +N organic soil displayed 11 open flowers throughout the experiment. We identified an interaction between bacteria and soil type for accumulated floral display (across three consecutive days; df = 327; $Z = 6.16$; $P < 0.001$; Figure 3C). Notably, NF+ plants in -N sand soil had an average of 30.5 times more flowers in 3 days compared to NF+ plants in +N organic soil. For NF- plants, we observed the production of 25 flowers under +N organic soil, while the NF- plants in -N sand soil produced only one flower during the experiment.

Finally, we detected an interaction between the presence of bacteria and soil type on floral density (df = 40.01; $F = 110$; $P < 0.001$; Figure 3D). NF+ plants in -N sand soil had a 14.5-fold higher floral density than on NF+ plants in +N organic soil. Calculating floral density for the NF- plants in -N sand soil was not feasible, because they only yielded a single flower throughout the experiment. In contrast, floral density remained lower within the NF- group in +N organic soil, 3.3 times less than the density on NF+ plants in +N organic soil and 48.3 times less than on NF+ plants in -N sand soil.

Effect of N-fixing bacteria and soil type on flowering time

We detected an interaction between bacterial presence and soil type on flowering time of *C. latistipula* (Figure 4). The presence of N-fixing bacteria led to earlier production of flowers with less variation in production among plants, but only in plants in -N sand soil (number of days until the first flower: df = 27, $Z = -10.81$, $P < 0.001$; Figure 4A). Similarly, the presence of bacteria led to an earlier flowering peak with less variation between plants but only in the -N sand soil (number of days until peak flowering: df = 27, $Z = -9.2$, $P < 0.001$; Figure 4B). In the other treatment combinations, flowering time was less predictable in its onset and peak production.

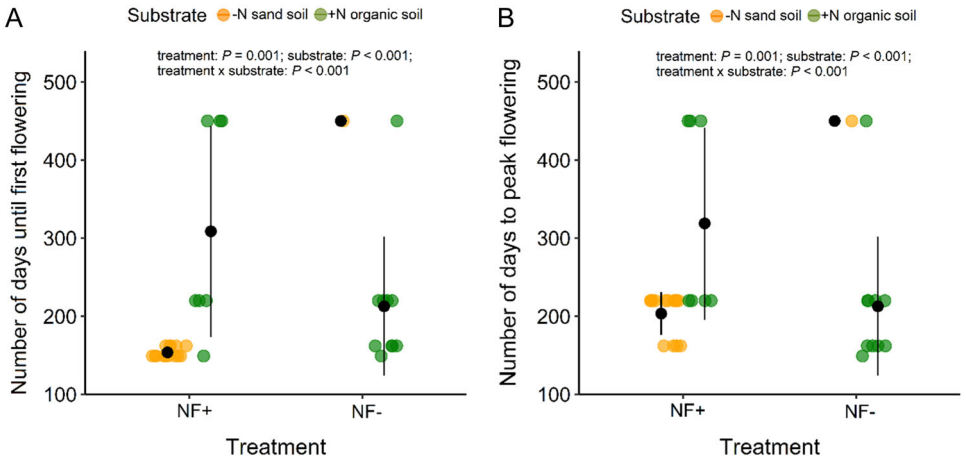


FIGURE 4 Flower phenology (onset, peak) for *Chamaecrista latistipula* plants with (NF+) and without (NF-) N-fixing bacteria in N-poor (-N) sand soil or N-rich (+N) organic soil. Flowering (A) onset and (B) peak. *P*-values were derived from generalized linear mixed models, reflecting the statistical significance of bacterial treatments, soil types, and their interaction on flower phenology metrics.

Effect of N-fixing bacteria and soil type on floral visual cues

The reflectance pattern and color vision model applied to petals and anthers in plants of *C. latistipula* did not differ greatly in the different treatment combinations (Appendices S3, S4). We did not detect any effect of bacteria and/or soil type on the green contrast of petals (df = 2, 18, $F = 1.9$, $P = 0.18$; Figure 5A) and the green contrasts of anthers (df = 2, 39, $F = 1.6$, $P = 0.21$; Figure 5B). We also did not find any effect of bacteria and/or soil type on the chromatic contrast of petals (df = 2, 18, $F = 0.87$, $P = 0.43$; Figure 5C). However, we did detect a difference in the chromatic contrast of anthers; NF+ plants (in -N sand soil or +N organic soil) had a higher chromatic contrast of anthers than NF- plants in +N organic soil (df = 2, 39, $F = 4.8$, $P = 0.014$; Figure 5D). Petal hue was also similar across treatments, whereas anther hue differed, but only in the anthers of NF+ plants grown in -N sand soil (Appendix S5).

DISCUSSION

Our findings underscore the impact of the interplay between N-fixing bacteria and soil nutrients on plant attributes associated with floral attractiveness to pollinators, improving on the knowledge about these effects available in the literature (see review in Table 1). Our results revealed the crucial role of N-fixing bacteria in determining plant attractiveness across various floral traits assessed, although these effects varied based on the soil conditions. In general, the flowers of plants in soil without bacteria (NF-) were significantly less attractive to pollinators, with almost no floral production, reduced color contrast for bee visual perception, lower plant positioning, and a delayed onset with inconsistent floral production over time. In comparison, the NF+ plants in -N sand soil inoculated with N-fixing bacteria had the most floral buds and flowers, with greater floral density, stronger color contrast for bee visual perception, higher positioning of flowers on plants, with early flower onset and constant production of flowers over time, showcasing the strong and positive impact of N-fixing bacteria on floral attractiveness to pollinators. However, NF+ plants in +N organic soil did not consistently meet our expectations of increased pollinator attractiveness and resembled NF- plants in +N organic soil. Our results underscore the pivotal role of the N-fixing bacteria in shaping multiple plant traits, including those linked to pollinator interactions.

High soil N is expected to reduce the resource allocation of plants to the N-fixing bacteria because plants can efficiently absorb N directly from the soil, which is energetically less costly than forming root nodules (Voisin et al., 2002). The lower demand for N fixation through nodules would potentially decrease the expression of nodulation genes or alter the signaling pathways (Ferguson and Mathesius, 2003; Andrews et al., 2011; Huisman and Geurts, 2020). Multiple edaphic factors, such as high N and organic matter content and soil compaction, may reduce root nodulation. Compacted soils with inadequate aeration can also reduce the species diversity of the N-fixing bacteria compared to sandy soils (Loureiro et al., 2007), whereas soils rich in organic matter can have positive or negative effects on the bacterial strains (e.g., Li et al., 2011). In our study, the relatively high amount of organic matter, with high availability of organic and inorganic N, and moderate degree of soil compaction, may have negatively impacted the bacteria in the +N organic soil, potentially decreasing root nodulation (<98% compared to NF+ plants in -N sand soil) because they did not need the bacteria to supply N. Although these plants seemed to be less dependent on N-fixing bacteria because of the low nodule production, they grew less and produced fewer flowers, so they apparently did not effectively use soil nutrients, especially N. If plants in N-rich organic soil cannot regulate nodule formation and fix N efficiently

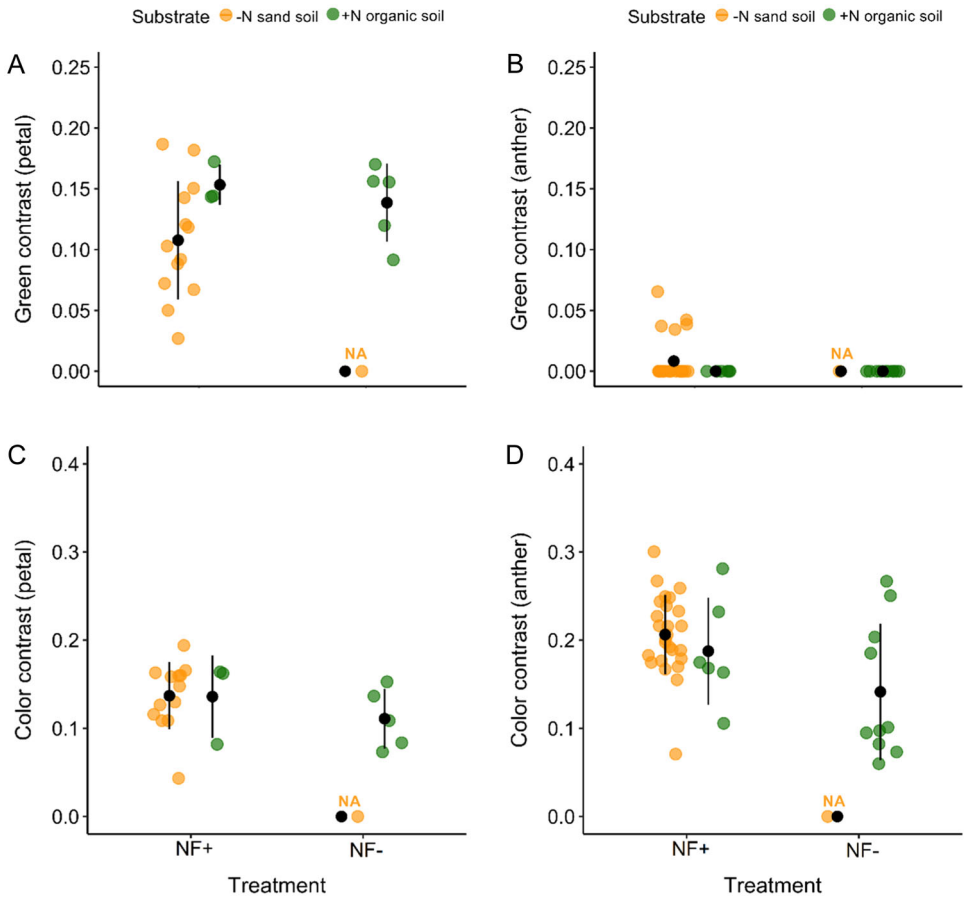


FIGURE 5 Green contrasts for (A) petals and (B) anthers and chromatic contrasts for (C) petals and (D) anthers of *Chamaecrista latistipula* plants. Given that we did not have any flowers in the NF- plants in the sand soil to estimate the contrasts of petals and anthers, we tested for direct differences between the three conditions with flowers available (NF+ plants in -N sand soil [control, natural field condition], NF+ plants in +N organic soil, NF- plants in +N organic soil) on green and chromatic contrasts for each flower part. The only difference detected was the chromatic contrast of anthers; anthers on NF+ plants (in -N sand soil or +N organic soil) had a higher chromatic contrast than those on NF- plants in +N organic soil (D). For details, see the full models in the Materials and Methods.

(Menge et al., 2015), they may encounter difficulties taking up adequate organic N directly from the soil, even when soil N is abundant. The mechanisms for this intriguing outcome remain unclear, indicating a need for additional experiments to unravel the edaphic factors and/or plant traits at play.

Variations in plant size and reproductive traits carry significant implications for all four plant treatment combinations in our study. Specifically, the control NF+ plants in -N sand soil were larger and produced more flowers, including more flower buds, more daily and cumulative flowers, and greater floral density compared to plants in the other three conditions. These attributes are commonly associated with enhanced attractiveness to pollinators (see Conner and Rush, 1996; Karron and Mitchell, 2012). Conversely, only one NF- plant in -N sand soil produced a flower throughout the experiment, reflecting extreme resource limitation. When faced with resource limitations, plants reallocate resources, often reducing floral production (Kuppler et al., 2021) or aborting initiated flowers (e.g., Scheepens et al., 2018; Cohen et al., 2021; Kuppler and Kotowska, 2021) in favor of vegetative expansion to enhance overall survival. The most intriguing findings concern the NF+ and NF- plants in +N organic soil; they all had similarly limited growth and reduced floral investment. However, NF+ plants in +N organic soil had superior growth and produced more flowers than the NF- plants did in the same soil. Thus, *C. latistipula* have some ability to utilize soil N, albeit to a minimal extent. Within the spectrum of plant dependency on N-fixing bacteria, *C. latistipula* aligns closely with the obligatory fixation strategies, though not as an extreme case as in other *Chamaecrista* species, such as *C. nictitans* (C. Souza et al., unpublished manuscript). This variability might be attributed to differing degrees of integration between bacterial strains and root nodules in *Chamaecrista* species, with shrubs and subshrubs in more open habitats achieving the highest integration (Naisbitt et al., 1992; Casaes et al., 2024). Some *Chamaecrista* species may form nodules without actively fixing N (de Faria et al., 2022; Casaes et al., 2024), possibly resembling more of a negotiation than unilateral

plant control (Menge et al., 2015). In our case, *C. latistipula* might have an additional intrinsic ability to inhibit nodule formation in N-rich soils despite the potential benefits of a bacterial symbiosis for plant growth and reproduction.

Previous studies have demonstrated that soil microbes associated with resource acquisition can influence plant life history traits (O'Brien et al., 2021), including flowering timing (O'Brien et al., 2021 and references therein). In our study, in -N sand soils where N availability is typically restricted, NF+ plants flowered earlier and reached peak flowering sooner, likely mirroring responses in the natural habitat of these plants in nutrient-poor sand with N-fixing bacteria. Floral production and flowering peak in these plants were also temporally more consistent, potentially indicating greater predictability for bees (see van der Kooi et al., 2021). Flowering synchrony increases the availability of plant mating partners. Conversely, both NF+ and NF- plants in +N organic soil initiated their first flowers, reaching peak flowering slightly later and more asynchronously. Effective pollination often hinges on phenological matching between plant flowering and pollinator activity, highlighting the importance of temporal synchrony (Rafferty et al., 2015).

Bee pollinators detect floral signals using three photoreceptors (UV, G, B) to discriminate colors effectively (Arnold and Chittka, 2019). Achromatic vision, mainly driven by the green photoreceptor, aids in long-distance perception, while chromatic vision, involving all photoreceptors, assists in short-distance perception (Giurfa et al., 1997; Giurfa and Lehrer, 2001; Brito et al., 2015). Our results showed that the effect of N-fixing bacteria on floral visual attractiveness is more pronounced in short-distance signals (chromatic contrast) than in long-distance signals (achromatic contrast). The anthers of NF+ plants had a higher chromatic contrast than those in NF- plants, and achromatic contrast was not affected. This result underscores how the absence of N-fixing bacterial nutrition can reduce bee visual perception of poricidal anthers, which is pivotal in attracting vibrating bee species to pollen flowers in buzz-pollination systems (De Luca and Vallejo-Marín, 2013; Saab et al., 2021; Delgado et al., 2023). Furthermore, the yellow and UV-absorbing color of anthers triggers innate responses in bees (Lunau, 1993; Lunau and Maier, 1995; Heuschen et al., 2005). Alterations in the visual patterns of anthers may affect bee responses (Lunau, 2000), potentially reducing their chances of locating these structures and subsequently releasing pollen grains by their vibrations. Changes in visual cues of anthers for bees may be linked with other unexplored floral traits, such as olfactory floral cues (Raguso and Willis, 2005), and need further investigation in this plant system.

CONCLUSIONS

Our findings elucidate the strong mutualistic relationship between *Chamaecrista latistipula* and N-fixing bacteria, diverging from typical expectations for a tropical legume species (Hedin et al., 2009; Menge et al., 2009). The higher abundance of large nodules per plant, particularly in the sand soil (-N sand soil), is evidence of the plant's reliance on these symbionts. This dependence is supported by the intimate ultrastructural integration between both partners, as seen by the formation of symbiosomes within root nodule cells densely infected with bacteroids (Naisbitt et al., 1992; de Faria et al., 2022; Casaes et al., 2024). Even in nutrient-rich organic soils, these plants exhibited an extreme reduction in floral attractiveness to pollinators, underscoring their reliance on N-fixing bacteria for nutrient acquisition and potentially other benefits. Importantly, our study reveals the critical gap in the literature connecting vital mutualisms in native ecosystems with crop production (see Table 1). Human activities have significantly increased soil N levels, especially from the excessive use of soil fertilizers (Vitousek et al., 1997). Still, this N increase can adversely affect the association between plants with a near-obligate fixation strategy, potentially compromising pollination services, as found here for *Chamaecrista* in +N organic soil. Future research should explore whether the pollen reward enclosed within poricidal anthers, accessible only to native bee species, is enriched in proteins/amino acids due to these vital bacteria partners in root nodules. Increased floral attractiveness may be linked to a higher quality or quantity of floral resources, influenced by the effectiveness of biological N fixation in these plants. Our results support that the interaction between aboveground and belowground mutualisms is context-dependent, with nutrient availability shaping resource allocation that connects belowground–aboveground mutualists.

AUTHOR CONTRIBUTIONS

C.S. conceived the idea for the article, analyzed the data, and led the writing of the manuscript. L.B.V.M. designed the study, collected data, and revised the manuscript. I.S.A. and P.J.B. assisted in interpreting the data and reviewed the manuscript. D.D.S. contributed to data manipulation and revision. A.N. conceptualized the study, analyzed and interpreted data, led the writing, and provided funding for the experiment. All authors contributed substantially to manuscript drafts and approved the final submission.

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DATA AVAILABILITY STATEMENT

The data are available on FigShare (doi:10.6084/m9.figshare.25727079).

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SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

APPENDIX S1. Synopsis of studies on the effects of belowground mutualists (N-fixing bacteria and/or mycorrhiza) on floral traits and/or pollination.

APPENDIX S2. Effect of N-fixing bacteria inoculation and soil types on the root nodule formation of *Chamaecrista latistipula*.

APPENDIX S3. Spectral reflectance of the leaves and floral parts of *Chamaecrista latistipula* in the four treatments.

APPENDIX S4. Bee hexagon visual model depicting the excitation (E) of ultraviolet (UV), green (G), and blue (B) photoreceptors of petals (left) and anthers (right).

APPENDIX S5. Hue angle (θ) for (A) petals and (B) anthers of *Chamaecrista latistipula* flowers.

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